

Tut.	Type	Tutorial
1		Aptitude Test
Unit : I – Fundamentals of Digital System		
2	A	Q-1 Perform the Following
		1. $(32.15)_{10} = ()_2$ 2. $(11011.101)_2 = ()_{10}$ 3. $(378.93)_{10} = ()_8$
		4. $(4057.06)_8 = ()_{10}$ 5. $(2598.675)_{10} = ()_{16}$ 6. $(110101.101010)_2 = ()_8$
		7. $(A0F9.0EB)_{16} = ()_{10}$ 8. $(367.52)_8 = ()_2$ 9. $(01011111011.011111)_2 = ()_{16}$
		10. $(3A9E.B0D)_{16} = ()_2$ 11. $(756.603)_8 = ()_{16}$ 12. $(B9F.AE)_{16} = ()_8$
	B	Q-2 Given that $(4096)_{10} = (1000)_x$, Find the value of x.
	A	Q-3 Convert the decimal number 250.5 to base 3, base 4, base 7 & base 8.
	C	Q-4 Covert Number $(4433)_5$ to Base 10 and Base 2.
	C	Q-5 Convert the decimal number $(2598.675)_{10}$ to Base 2, Base 4, Base 8, Base 16.
3	B	Q-1 Convert $(33.45)_{10}$ to binary. Result should be accurate to within $(0.01)_{10}$.
	B	Q-2 Convert $(99.99)_{10}$ to Octal. Result should be accurate to within $(0.02)_{10}$.
	C	Q-3 Perform the following for Octal numbers. 1. $26.5 * 2.5$ 2. Divide 153.6 by 7
	C	Q-4 Perform the following for Hexadecimal numbers. 1. $92.5 * B.3$ 2. Divide 6ABF.6D by 1A
	B	Q-5 Perform the following; 1. $(-128)_{10} = ()_2$ 2. $(-1)_{10} = ()_2$ 3. $(-39.79)_{10} = ()_2$ 4. $(-49.02)_{10} = ()_2$
	C	Q-6 Find the value of X. 1. $\sqrt{(61)_x} = (7)_{10}$ 2. $\sqrt{(144)_x} = (18)_{10}$



	A	Q-9 Reduce the Expression: $F = \prod M(2,8,9,10,11,12,14)$
	B	Q-10 Reduce the Expression: $F = \prod M(0,1,3,4,5,6,7,13,15)$
6	B	Q-1 Reduce using K-Map: $f = \sum m(0, 2, 4, 6, 7, 8, 10, 12, 13, 15)$
	C	Q-2 Obtain the simplified expression using K-Map: $F = ABD + A'C'D' + A'B + A'CD' + AB'D$
	A	Q-3 Minimize following Boolean function using K-Map: $F = \sum m(1,2,4,6,7,11,15)+d(0,3)$
	B	Q-4 Simplify following Boolean function using K-Map: $F(W,X,Y,Z) = \sum m(1,3,5,8,9,11,15)+d(2,13)$
	B	Q-5 Reduce the expression in SOP and POS form using K-Map: $F(A,B,C,D) = \sum m(1,5,6,12,13,14)+d(2,4)$
	C	Q-6 Reduce using K-Map: $f = \sum m(6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$
	C	Q-7 Reduce using K-Map: $f = \sum m(0, 1, 4, 5, 6, 7, 9, 11, 15) + d(10, 14)$
7	A	Q-1 Using D as MEV, reduce $Y = A'B'C'D' + A'B'CD' + AB'C'D' + AB'C'D + AB'CD + AB'CD'$
	A	Q-2 Simplify the following Boolean function using VEM. $F = AB'CD + A'BC'D + AB'CD' + A'B'C'D$
	B	Q-3 Simplify following Boolean function using VEM. $F = A'B'C'D + A'BC'D' + A'BC'D + AB'C'D' + AB'CD' + AB'CD + ABCD'$
	B	Q-4 Using E as the MEV, reduce $F = A'B'C'D'E + A'B'C'DE + A'BCD'E' + A'BCD'E + AB'C'D'E' + AB'C'D'E + AB'C'D + A'BCDE'$
	C	Q-5 Using C & D as MEVs, reduce $Y = A'B'C'D' + A'B'C'D + A'B'CD' + A'BCD + ABCD + AB'CD$
	C	Q-6 Simplify following Boolean function using VEM. $F(A,B,C,D) = \sum m(1,2,3,5,6,7,8,9,12,13,15)$
	C	Q-7 Simplify following Boolean function using VEM. $F(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$
8	A	Q-1 Simplify the following Boolean function by means of Tabulation method: $F(A,B,C,D) = \sum m(1,2,3,5,6,7,8,9,12,13,15)$
	B	Q-2 Simplify the following Boolean function by means of Tabulation method: $F(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$

	C	Q-3 Simplify the following Boolean function by means of Tabulation method: $F = A'B'C'D + A'BC'D' + A'BC'D + AB'C'D' + AB'CD' + AB'CD + ABCD'$
	C	Q-4 Simplify the following Boolean function by means of Tabulation method: $F = \sum_m (0, 2, 4, 6, 7, 8, 10, 12, 13, 15)$
Unit : III – Combinational Digital Circuits		
9	A	Q-1 Minimize the logic function $X = A (B'+C') (A+D)$. Also realize the reduced function using NOR gates only.
	B	Q-2 Simplify using Boolean laws and draw the logic diagram for the given expression. $F = (ABC)' + (AB)'C + A'BC' + A(BC)' + AB'C$
	A	Q-3 Reduce the expression $F = \sum_m (0,2,3,4,5,6)$ using K-map and implement using NAND gates only.
	A	Q-4 Implement following Boolean function using 8 : 1 multiplexer. $Y (A,B,C,D) = \sum_m (15,7,4,6,8,9,12,14)$
	B	Q-5 Implement following Boolean function using 8 : 1 multiplexer. $F (A,B,C,D) = \sum_m (2,3,5,7,8,9,12,13,14,15)$
	A	Q-6 Implement following Boolean function using 4 : 1 multiplexer. $F (A,B,C) = \sum_m (1,3,5,6)$
	B	Q-7 Implement following Boolean function using 4 : 1 multiplexer. $F (A,B,C) = \sum_m (1,2,4,7)$
	C	Q-8 Design a 4-Bit Binary to BCD Code Converter.
10		Perform hands on practical of logic gate.
11		Perform hands on practical of logic circuits.
Unit : IV – Sequential Digital Circuits		
12	A	Q-1 Design 3-bit ripple up-counter using negative edge JK Flip-flops.
	B	Q-2 Design 4-bit ripple up-counter using positive edge JK Flip-flops.
	B	Q-3 Design 3-bit ripple Down-counter using negative edge T Flip-flops.
	C	Q-4 Design 4-bit ripple Down-counter using positive edge T Flip-flops.
13	C	Q-1 Design Modulo-8 counter using T Flip-flops.

	A	Q-2 Design Mod-10 (Decade) Counter using T Flip-flops.
	B	Q-3 Design Mod-12 using JK Flip-flops.
14	A	Q-1 Design 3-bit synchronous up counter using T Flip-flop.
	B	Q-2 Design a synchronous BCD counter with JK Flip-flop.
15	A	Q-1 Design a counter to generate the repetitive sequence 0, 3, 5, 7, 4 using D Flip-flop.
	B	Q-2 Design a counter to generate the repetitive sequence 0, 1, 2, 4, 3, 6 using T Flip-flop.
	C	Q-3 Design a JK counter that goes through states 3,4,6,7 and 3.